

Notes: Stiglitz and Weiss (AER, 1981)

Credit Rationing in Markets with Imperfect Information

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model setup and key payoffs	4
3.1	Borrowers, projects, and limited liability	4
3.2	Riskiness and mean-preserving spreads	4
3.3	Definition of credit rationing	4
4	Interest rate as a screening device	5
4.1	Adverse selection through borrower application decisions	5
4.2	Why expected bank return can be nonmonotonic	5
4.3	Credit rationing equilibrium	6
4.4	Multiple modes and two-rate equilibria	6
5	Interest rate as an incentive mechanism	6
5.1	Risk shifting through project choice	6
5.2	Two-project example	7
5.3	Relation to credit rationing	7
6	Collateral, limited liability, and redlining	7
6.1	Why collateral need not clear the market	7
6.2	Collateral as a screening device	7
6.3	Observationally distinguishable borrowers and redlining	8
6.4	Debt versus equity and principal-agent contracts	8
7	Figures	9
7.1	Figure 1 and Figure 2a: Bank return and borrower convexity	9
7.2	Figure 2b and Figure 3: Bank concavity and adverse selection	9
7.3	Figure 4 and Figure 5: Rationing equilibrium and multiple modes	10
7.4	Figure 6 and Figure 7: Incentives and collateral screening	11
7.5	Figure 8 and Figure 9: Collateral and redlining	12
7.6	Figures 10a and 10b: Principal-agent interpretation	13

8	Short summary	13
9	Conclusion	14
9.1	Core conclusion	14
9.2	Economic intuition	14
9.3	Incremental contribution revisited	14
9.4	Implications	14
9.5	Limitations	15
9.6	Takeaway	15

1 Big picture

- **Research question.** Why can credit be rationed in equilibrium? More specifically, why might banks refuse loans to some borrowers even when those borrowers are willing to pay a higher interest rate or pledge more collateral?
- **Main idea.** In loan markets, the interest rate is not just a price. It also changes the composition of borrowers and the behavior of borrowers. When the price affects the quality of the transaction, the market-clearing price need not be the equilibrium price.
- **Two central mechanisms.** Higher interest rates can create **adverse selection** by driving safer borrowers out of the applicant pool. Higher interest rates can also create **moral hazard** by inducing borrowers to choose riskier projects.
- **Bank objective.** A bank cares about the expected return on its loan portfolio, not only about the promised interest rate. If a higher promised rate attracts or induces riskier loans, the bank's expected return may fall.
- **Credit rationing equilibrium.** The bank-optimal interest rate can be below the interest rate that would equate loan demand and loan supply. At that rate, demand for loans exceeds supply, but banks do not raise the rate because doing so would lower expected profits.
- **Collateral is not a complete solution.** Increasing collateral requirements can also affect selection and incentives. It may attract wealthier but less risk-averse borrowers or induce riskier actions, so it need not clear the market.
- **Redlining implication.** Some observable borrower groups may be denied credit at all interest rates if the bank's maximum expected return on loans to that group is too low relative to the cost of funds or relative to other borrower groups.
- **Economic takeaway.** The paper overturns the simple view that competitive prices always clear markets. If prices screen agents or change their actions, equilibrium can involve queues, rationing, and non-market-clearing prices.

2 Incremental contribution of the paper

- **First theoretical justification of true credit rationing.** Earlier explanations often relied on disequilibrium, usury laws, or upward-sloping loan supply schedules. Stiglitz and Weiss show that true rationing can arise in equilibrium under imperfect information.
- **Distinguishes promised return from expected return.** The paper emphasizes that a higher contractual interest rate does not necessarily imply a higher expected return to the lender.
- **Makes adverse selection endogenous to the interest rate.** The willingness to pay a high rate reveals borrower type. Borrowers who accept high rates may be worse risks because they expect to default with higher probability.
- **Adds borrower incentive effects.** The interest rate changes project choice. Because borrowers have limited liability, high rates make high-upside, low-success-probability projects more attractive.

- **Explains why collateral may not eliminate rationing.** Collateral improves recovery from any given borrower, but it can also worsen the pool of borrowers or change borrower behavior.
- **Links credit markets to principal-agent theory.** The model applies more broadly to settings in which fixed payments, default, effort, and risk choices matter, such as landlord-tenant or employer-worker relationships.
- **Policy relevance.** Monetary policy can affect investment through credit availability rather than only through interest rates. Legal restrictions such as usury ceilings or other interventions may improve allocation in some cases, although the paper does not argue that rationing always occurs.

3 Model setup and key payoffs

3.1 Borrowers, projects, and limited liability

The model considers many banks and many potential borrowers. Borrowers need a loan of size B to undertake a project. A project produces a random gross return R . Borrowers can post collateral C , but they are protected by limited liability.

For a loan with interest rate r , the borrower's net payoff is

$$\pi(R, r) = \max\{R - (1 + r)B, -C\}.$$

The bank's payoff is

$$\rho(R, r) = \min\{R + C, B(1 + r)\}.$$

Interpretation. The borrower payoff is convex in project returns because downside losses are capped by collateral and upside gains remain with the borrower. The bank payoff is concave because the bank gets the promised repayment in good states but bears default losses in bad states.

3.2 Riskiness and mean-preserving spreads

Borrowers can differ in the riskiness of their projects. The paper denotes project return distributions by $F(R, \theta)$, where a higher θ corresponds to greater risk in the sense of a mean-preserving spread.

- Projects can have the same mean return but different risk.
- Riskier projects are more attractive to borrowers because borrower payoffs are convex.
- Riskier projects are less attractive to banks because bank payoffs are concave.

Interpretation. This wedge between borrower preferences and bank preferences is the basic friction. Borrowers like upside risk because they do not bear the full downside. Banks dislike the same risk because default cuts off their contractual claim.

3.3 Definition of credit rationing

The paper uses a strong definition of credit rationing. Rationing exists when either:

1. among loan applicants who appear identical to the bank, some receive loans and others do not, and rejected applicants cannot obtain loans by offering to pay a higher interest rate; or
2. identifiable borrower groups cannot obtain loans at any interest rate under a given supply of credit, even though with a larger supply of credit they would receive loans.

Interpretation. This is not just an upward-sloping interest schedule or a temporary shortage. It is an equilibrium allocation in which quantity, not price, determines who receives funds.

4 Interest rate as a screening device

4.1 Adverse selection through borrower application decisions

The first mechanism is adverse selection. For a given interest rate, there is a critical risk level such that only borrowers above that risk level apply for loans.

- Low-risk borrowers are less willing to accept high interest rates because they expect to repay more often.
- High-risk borrowers are more willing to accept high interest rates because default is more likely.
- Raising the interest rate can therefore remove safer borrowers from the pool and leave a riskier applicant pool.

The paper's logic can be summarized as:

$$\text{higher } r \Rightarrow \text{higher cutoff risk among applicants} \Rightarrow \text{riskier loan pool.}$$

Economic message. A borrower's willingness to pay a high interest rate is informative. It is not necessarily good news for the bank; it can signal that the borrower is a bad risk.

4.2 Why expected bank return can be nonmonotonic

Let $\bar{\rho}(r)$ denote the bank's expected return per dollar loaned at interest rate r . There are two opposing effects of raising r :

- **Direct price effect.** Holding borrower composition fixed, a higher interest rate raises the promised repayment.
- **Adverse-selection effect.** A higher interest rate worsens the borrower pool by driving out safer borrowers.

If the adverse-selection effect is strong enough, $\bar{\rho}(r)$ first rises and then falls. The bank then has an interior optimal interest rate r^* .

Interpretation. The bank does not maximize the contractual rate. It maximizes expected return. The two differ because the contractual rate changes who borrows.

4.3 Credit rationing equilibrium

If the demand for loans at r^* exceeds the supply of funds available to banks, a rationing equilibrium can arise.

- Borrowers want more loans at the bank-optimal rate.
- A higher rate would reduce loan demand, but it would also lower the expected return to the bank.
- Banks therefore refuse to raise rates.
- Some observationally identical applicants receive loans, while others do not.

Economic message. The market-clearing rate is not necessarily an equilibrium rate. A bank may prefer a lower rate with excess demand because that lower rate gives a better applicant pool.

4.4 Multiple modes and two-rate equilibria

The paper also allows the expected-return function to have multiple local maxima. In this case, equilibrium can involve either a single interest rate or two interest rates.

- At the lower rate, there can be excess demand for credit.
- Some rejected borrowers may obtain loans at a higher rate.
- Banks must earn the same expected return on the loans they make at the different rates.

Interpretation. Multiple rates can coexist because rates are associated with different borrower pools. The rate is partly a screening device, not just a uniform price of funds.

5 Interest rate as an incentive mechanism

5.1 Risk shifting through project choice

The second mechanism is incentive-based. Suppose a borrower can choose among projects after receiving the loan. The bank cannot perfectly monitor or contract on that choice.

- A high interest rate reduces the borrower's payoff in successful states.
- Because the borrower keeps only returns above the debt repayment and has limited liability in bad states, higher rates make upside-skewed projects relatively more attractive.
- Thus, a higher interest rate can induce borrowers to switch from safer projects to riskier projects.

If a borrower is initially indifferent between two projects, an increase in r makes the borrower prefer the project with the higher probability of bankruptcy.

Economic message. Interest rates affect actions, not just participation. The same borrower may become riskier when faced with a higher rate.

5.2 Two-project example

The paper gives a simple example with two projects:

- Project b is safer: lower payoff if successful but higher probability of success.
- Project a is riskier: higher payoff if successful but lower probability of success.

At the interest rate where the borrower is indifferent,

$$[R^a - (1 + r)B]p^a = [R^b - (1 + r)B]p^b.$$

Below the cutoff rate, the borrower chooses the safer project. Above it, the borrower chooses the riskier project. If the switch is sufficiently harmful to the bank, the bank's expected return falls when the interest rate rises above the cutoff.

Interpretation. The model captures a standard debt-overhang or risk-shifting force. The more debt-like the borrower payoff, the stronger the incentive to gamble.

5.3 Relation to credit rationing

The incentive mechanism alone can generate a bank-optimal interest rate below the market-clearing rate. Even if borrower composition were fixed, the bank may avoid raising the rate because doing so changes borrower behavior.

Economic message. Rationing does not require that different borrowers have different hidden types. It can also arise when the same borrowers change their actions in response to contract terms.

6 Collateral, limited liability, and redlining

6.1 Why collateral need not clear the market

A natural objection is that banks could clear excess demand by requiring more collateral. The paper argues that this need not work.

- For any given borrower and project, more collateral improves the bank's recovery in bad states.
- But collateral requirements also change who borrows and what borrowers do.
- If higher-collateral borrowers are wealthier and less risk averse, they may be more willing to undertake risky projects.
- In a multiperiod setting, tighter initial funding can induce firms to take actions that raise the need for future risky financing.

Interpretation. Collateral has a beneficial direct effect and a potentially harmful selection or incentive effect. The net effect can be negative for the bank.

6.2 Collateral as a screening device

The paper develops a model in which all borrowers have the same utility function but differ in wealth. With decreasing absolute risk aversion, wealthier borrowers are less risk averse.

- Only borrowers with sufficient wealth can post high collateral.
- Among borrowers who apply, wealthier borrowers may choose riskier projects.
- Raising collateral can therefore select toward a riskier pool.

The key tension is:

more collateral $\Rightarrow$ higher recovery from a fixed borrower

but also $\Rightarrow$ riskier marginal borrower and possibly lower expected return.

Economic message. Collateral is not a neutral price either. Like the interest rate, it screens borrowers and can affect actions.

6.3 Observationally distinguishable borrowers and redlining

The paper extends the analysis to borrower groups that banks can distinguish. Each group has its own expected-return function, $\rho_i(r_i)$.

- Banks lend to the groups that offer the highest expected return per dollar.
- If the maximum expected return on a group is below the cost of funds, that group receives no loans.
- Such a group is effectively redlined: it cannot obtain credit at any interest rate under the prevailing supply conditions.

Important implication. Redlined groups may still have projects with high total social returns. The problem is that banks cannot capture those returns or cannot sufficiently separate good projects from bad projects within the group.

6.4 Debt versus equity and principal-agent contracts

The paper then broadens the argument. In any principal-agent setting, contract terms can affect selection and incentives.

- Pure equity or share contracts can weaken effort incentives because the agent receives only part of the marginal product.
- Pure debt or fixed-rent contracts impose default risk and can induce risk shifting.
- Nonlinear contracts, collateral, monitoring, and contingent future credit can help, but they do not necessarily eliminate rationing.

Interpretation. Credit rationing is one example of a broader phenomenon: when contracts cannot fully condition on actions and states, the equilibrium contract may not clear the market.

7 Figures

7.1 Figure 1 and Figure 2a: Bank return and borrower convexity

Read Figure 1:

- The expected return to the bank rises with the interest rate at first and then falls.
- The peak is the bank-optimal rate r^* .
- Beyond r^* , higher rates worsen selection or incentives enough to reduce expected return.

Read Figure 2a:

- Borrower profits are a convex function of project returns.
- Limited liability caps downside losses at collateral, while upside gains remain with the borrower.
- This convexity makes riskier projects attractive to borrowers.

Economic message: The central friction is the mismatch between borrower and lender payoffs. Borrowers like upside risk; banks worry about repayment.

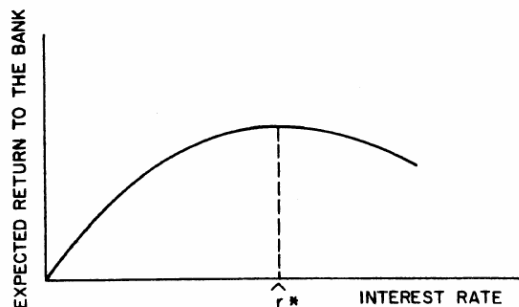


Figure 1: Expected bank return is maximized at an interior interest rate.

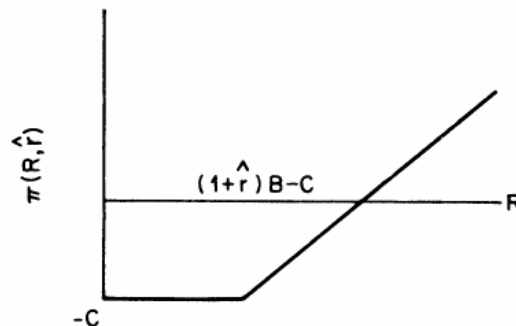


Figure 2a: Borrower payoff is convex in project returns because of limited liability.

7.2 Figure 2b and Figure 3: Bank concavity and adverse selection

Read Figure 2b:

- The bank's payoff is concave in project returns.
- Once the loan is repaid, the bank does not share additional upside.
- In bad states, the bank bears default losses.

Read Figure 3:

- Raising the interest rate can cause low-risk borrowers to drop out.
- At lower rates, both types apply.
- At higher rates, only high-risk borrowers apply.

Economic message: The interest rate acts as a screening device. High rates attract or retain worse risks.

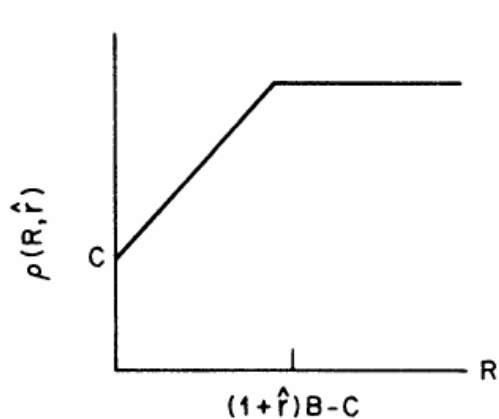


Figure 2b: Bank payoff is concave in project returns.

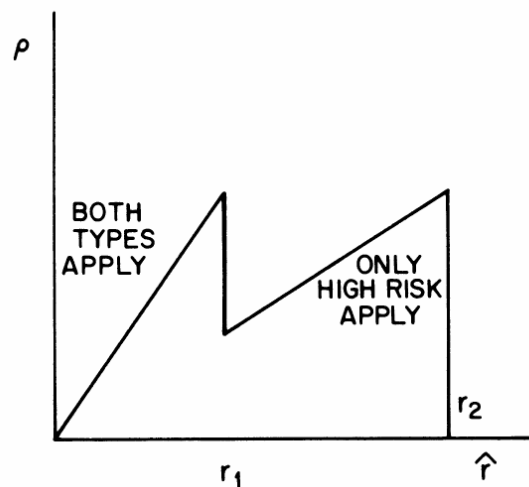


Figure 3: As the interest rate rises, safer borrowers may leave the applicant pool.

7.3 Figure 4 and Figure 5: Rationing equilibrium and multiple modes

Read Figure 4:

- Loan demand depends on the interest rate charged to borrowers.
- Loan supply depends on the expected return banks can offer depositors.
- Because expected bank return is nonmonotonic in the loan rate, the conventional supply-demand diagram does not apply.
- The bank-optimal rate can generate excess demand for credit.

Read Figure 5:

- The expected-return schedule can have multiple local maxima.
- This can support equilibria with more than one loan rate.
- There may still be excess demand at the lower rate.

Economic message: The equilibrium rate is chosen to maximize expected bank return, not to clear the market.

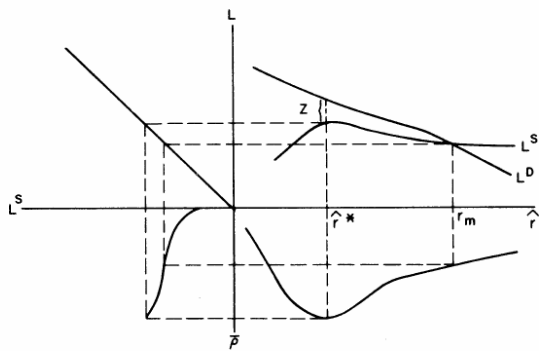


Figure 4: Determination of a credit-rationing market equilibrium.

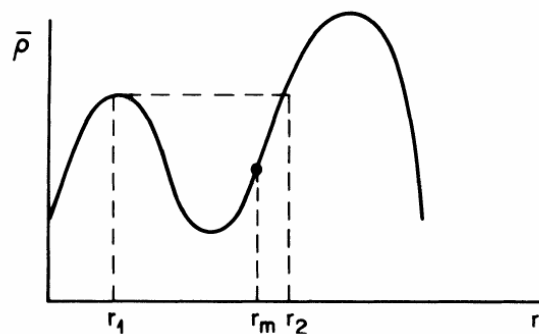


Figure 5: A two-interest-rate equilibrium when expected return has multiple modes.

7.4 Figure 6 and Figure 7: Incentives and collateral screening

Read Figure 6:

- Below the cutoff rate, the borrower chooses the safer project.
- Above the cutoff rate, the borrower chooses the riskier project.
- The bank's expected return can fall once the interest rate induces risk shifting.

Read Figure 7:

- Collateral affects which wealth types borrow and how they invest.
- In the middle wealth range, borrowers apply for loans.
- Wealthier borrowers may self-finance and choose risky projects.

Economic message: Contract terms affect behavior and selection at the same time. Interest rates and collateral both have screening and incentive effects.

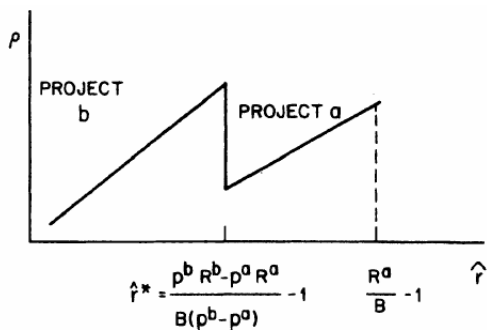


Figure 6: Above the cutoff interest rate, the risky project is chosen and bank return falls.

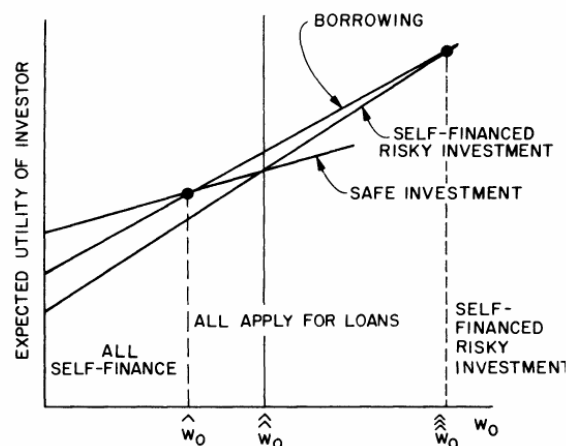


Figure 7: Collateral serves as a screening device across wealth types.

7.5 Figure 8 and Figure 9: Collateral and redlining

Read Figure 8:

- Increasing collateral raises bank return from a fixed borrower at first.
- At a threshold, however, the applicant pool changes discontinuously.
- The adverse selection effect can dominate the direct collateral effect.

Read Figure 9:

- Different observable borrower groups have different expected-return schedules.
- Some groups receive credit; others may be excluded entirely.
- This is the paper's redlining result.

Economic message: Rationing can occur within an observationally identical group or across distinguishable groups.

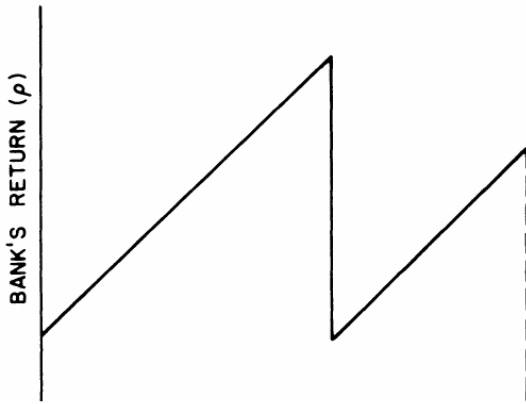


Figure 8: Increasing collateral can lower the bank's expected return.

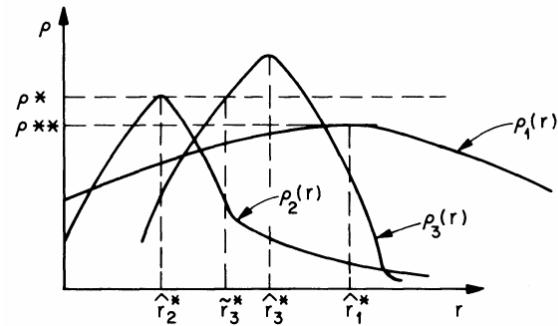


Figure 9: If borrower groups differ, some groups may be redlined.

7.6 Figures 10a and 10b: Principal-agent interpretation

Read Figures 10a and 10b:

- Figure 10a shows the tenant or agent's payoff under a fixed-rent type contract.
- Figure 10b shows the landlord or principal's payoff.
- The same payoff geometry applies to debt contracts: the agent keeps upside beyond the fixed payment, while the principal's payoff is capped.

Economic message: The theory is not only about banks. It applies to any contract in which fixed payments, default, unobserved actions, and risk choices interact.

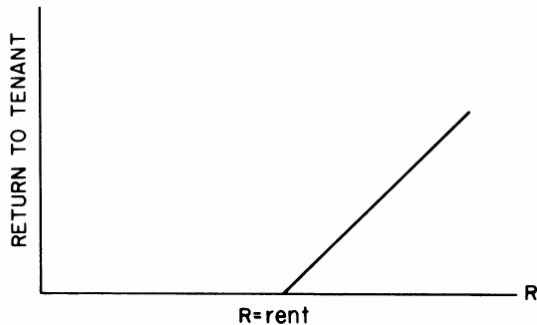


Figure 10a: Return to the tenant or agent under a fixed-rent contract.

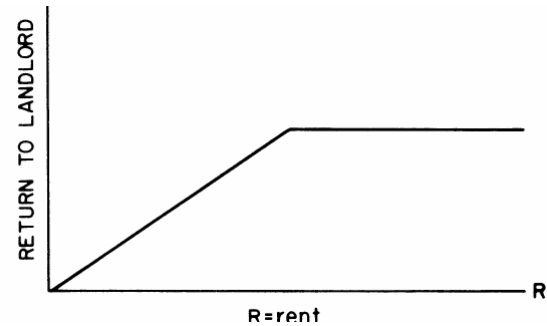


Figure 10b: Return to the landlord or principal under the same contract.

8 Short summary

- The paper explains how credit rationing can occur in equilibrium, not only in disequilibrium or because of legal restrictions.

- The key reason is that loan terms affect the quality of the loan.
- Higher interest rates can worsen adverse selection by attracting riskier borrowers or driving safer borrowers out.
- Higher interest rates can also worsen incentives by inducing borrowers to choose riskier projects.
- Because borrower payoffs are convex and bank payoffs are concave, borrowers and banks rank project risk differently.
- The bank's expected return can be nonmonotonic in the interest rate.
- If the bank-optimal interest rate is below the market-clearing rate, banks ration credit rather than raise rates.
- Collateral is also not a guaranteed solution because it can worsen selection or incentives.
- Some borrower groups may be redlined if banks cannot earn enough expected return from lending to them.
- The broader message is that prices do not necessarily clear markets when prices have sorting and incentive effects.

9 Conclusion

9.1 Core conclusion

Stiglitz and Weiss show that credit rationing can be an equilibrium outcome in competitive loan markets with imperfect information. Banks may refuse loans even when rejected borrowers are willing to pay higher interest rates because higher rates can lower expected bank returns.

9.2 Economic intuition

The intuition comes from the payoff structure of debt. Borrowers keep upside gains but have limited downside losses. Banks receive a capped repayment and bear default losses. This makes risk attractive to borrowers and costly to banks. Interest rates and collateral therefore affect not only loan demand but also borrower quality and borrower behavior.

9.3 Incremental contribution revisited

The paper's contribution is to replace a mechanical price-clearing view of credit markets with an information-based contract view. In this view, loan terms screen hidden types and shape hidden actions. The equilibrium contract is chosen for expected profitability, not for market clearing.

9.4 Implications

The findings imply that the quantity of credit can matter independently of interest rates. Monetary policy, banking regulation, and financial intermediation can affect real investment by changing access to credit. The model also gives a theoretical foundation for why some groups or firms may face persistent financing constraints.

9.5 Limitations

The paper is theoretical and stylized. It does not estimate the frequency or magnitude of credit rationing in real loan markets. The model abstracts from many features of modern banking, including relationship lending, securitization, credit scoring, renegotiation, and dynamic reputation. Still, the core logic remains central: loan prices can affect loan quality.

9.6 Takeaway

The enduring takeaway is that the Law of Supply and Demand is not automatic. It follows only when prices do not change the composition or actions of market participants. In credit markets with imperfect information, interest rates and collateral affect borrower quality and borrower behavior, so equilibrium can involve rationing.